

Rajagopal Somaskandan

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SUMMARY

Entry-level Machine Learning Engineer with hands-on experience building end-to-end AI systems using FastAPI, PyTorch, and modern ML frameworks. Built production-oriented projects spanning fraud detection, recommendation systems, computer vision, and privacy-preserving crowd analytics. Strong interest in backend ML systems, explainable AI, A/B experimentation, and scalable deployment.

EDUCATION

Sathyabama Institute of Science and Technology <i>M.Sc. Computer Science</i>	Expected 2027
Sathyabama Institute of Science and Technology <i>B.Sc. Computer Science (AI Specialization)</i>	2025

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL
Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, A/B Experimentation, Explainable AI (SHAP)
Deep Learning: PyTorch, Neural Networks, CNNs, RNNs, LSTMs, LLM Fine-Tuning (LLaMA)
NLP & Retrieval: Transformers, Embeddings, RAG, BERT, Sentence-Transformers, FAISS, ChromaDB
Backend & APIs: FastAPI, REST APIs
Data Pipelines & ETL: Selenium, Java-based scrapers
Computer Vision & 3D: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting, FFmpeg
Databases: PostgreSQL, MongoDB
MLOps & Deployment: Docker, Git, GitHub, CI/CD (GitHub Actions), Netlify, Render, ngrok
Frontend: React, Tailwind CSS, Streamlit

EXPERIENCE

Production AI Developer (Intern) <i>InnoXR Labs Pvt. Ltd.</i>	Mar 2026 – Jun 2026 <i>Chennai, India</i>
- Built and deployed ML-powered FastAPI microservices for real-time inference in AR and immersive applications. - Designed API-first ML systems integrating model inference with scalable backend endpoints; managed deployment via Docker and ngrok tunneling. - Delivered end-to-end features from model development to production deployment in a fast-paced startup environment.	
Data Analyst Intern <i>India Advocacy</i>	Jul 2024 – Oct 2024 <i>Chennai, India</i>
Built Java and Selenium ETL pipelines to scrape and process 1M+ records, transforming raw data into 15–20k structured datasets through multi-stage validation.	
Security Trooper (National Service) <i>Singapore Armed Forces</i>	Jan 2020 – Jan 2022 <i>Singapore</i>
- Executed security and vehicular operations in regulated environments, ensuring strict protocol compliance and operational readiness. - Supported COVID-19 response initiatives through structured data collection and reporting under time-sensitive conditions.	
Sales Associate (Part-Time) <i>Bata</i>	Feb 2022 – Aug 2022 <i>Singapore</i>
- Handled high-volume customer interactions in a fast-paced retail environment, ensuring efficient service delivery. - Managed billing, inventory coordination, and issue resolution to support smooth store operations.	

PROJECTS

FinGuard Pro: Real-Time Fraud Detection & Compliance System	IEEE ICAAIC 2025
- Built a UPI fraud detection system trained on 100,000 synthetic transactions using 25+ engineered features including geo-deviation scores and amount-to-average ratios. - Conducted A/B experimentation across XGBoost, LightGBM, and CatBoost with Optuna TPE hyperparameter optimization; XGBoost selected (94% recall, ROC-AUC 0.9951, F1 0.57) over LightGBM (F1 0.56) and CatBoost (F1 0.48). - Integrated SHAP TreeExplainer for global/local interpretability and a fuzzy AML screening engine (RapidFuzz, 5,000-entity watchlist) with rule-based detection of structuring, round-tripping, and velocity anomalies. - Deployed a dual-dashboard (FastAPI + Streamlit) with real-time alerts, SHAP visualizations, and automated PDF/ZIP compliance report generation; inference latency under 3 seconds per transaction.	
CareerGenie: AI Career Platform	Hackathon Winner, Industry Innovation '26
- Built an end-to-end RAG pipeline using MiniLM embeddings and ChromaDB for semantic job matching, replacing keyword-based retrieval with skill-aware vector search. - Designed a multi-agent orchestration system (ReAct-style) with 5 specialized agents (Resume, Job, Interview, Roadmap, Advisor) and an Agent Debate System (3 proposers, critic, synthesizer) to reduce hallucinations. - Engineered a Learning-to-Rank engine using BPR trained on click, application, and offer signals; integrated SerpAPI + Google Trends with weighted scoring (semantic 35%, skills 45%, title 20%). - Performed LLM fine-tuning on LLaMA via multi-provider waterfall (Groq → Anthropic) improving response quality and reducing latency under degraded provider conditions. - Delivered full-stack system: React + Tailwind frontend deployed on Netlify, FastAPI backend via ngrok tunneling, supporting ATS scoring, resume rewriting, mock interviews, and learning roadmaps.	
MonoSplat: Single-Video 3D Reconstruction Pipeline	Internship Project, 2026
- Engineered a 3D reconstruction pipeline using COLMAP and Gaussian Splatting in PyTorch. - Designed a hybrid CPU-GPU data pipeline, optimizing preprocessing with FFmpeg for cloud execution. - Built an async FastAPI backend with a Three.js viewer for interactive 3D visualization. - Focused on performance and usability, enabling smooth rendering and faster end-to-end processing.	
S.A.F.E: Privacy-Preserving Crowd Anomaly Detection	Design Thinking Project, 2026
- Developed a privacy-first crowd anomaly detection system using Farneback optical flow and a 2×3 zone grid to extract density, velocity, and direction variance metrics without tracking individuals. - Conducted A/B experimentation across 5 unsupervised models (MAD, Z-Score, One-Class SVM, Isolation Forest, LSTM Autoencoder) trained on 15,928 zone records (ETH/UCY); validated on 13,805 records (Zara1/Zara2). MAD selected as primary model after cross-domain evaluation on the Mall dataset (ROC-AUC 0.8391, F1 0.4130); reduced false positives by 40–60% via adaptive camera shake detection (threshold 0.25, 3-frame check). - Built a FastAPI backend and Streamlit dashboard with explainable real-time alerts and zone-level reasoning (e.g., “Density >50% in Zone 3”) for safety operators.	
SourceUp: Explainable Supplier Recommendation System	Final Year Capstone, 2026
- Implemented semantic supplier search using Sentence-BERT and FAISS for natural language queries. - Designed a ranking model using Scikit-learn with cost, delivery time, and reliability constraints. - Added an explainability layer to provide feature-level insights for supplier selection and risk assessment.	

PUBLICATIONS & CONFERENCES

FinGuard Pro: Explainable AI for Financial Fraud Detection <i>ICAAIC 2025</i>	ISBN: 979-8-3315-6587-9
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ACHIEVEMENTS & HACKATHONS

Industry Innovation Hackathon <i>Sathyabama Institute of Science and Technology – National Level</i>	Mar 2026
Won Best Industry Innovation Award at Industry Innovation '26.	
Origin 24-Hour Hackathon <i>SIMATS Engineering, Chennai</i>	Apr 2026
Placed 4th at Origin 24-Hour Hackathon.	

CERTIFICATIONS

AI & ML <i>Space Zee Technologies</i>	Jul 2024
Python Core <i>Space Zee Technologies</i>	Feb 2024